

School of Mathematics and Statistics

MT5864 Advanced Group Theory

Problem Sheet IV: Semidirect Products

1. Let p be a prime number. Let $N = C_p \times C_p \times \cdots \times C_p$ be a direct product of n copies of a cyclic group of order p . Show that

$$\text{Aut } N \cong \text{GL}_n(\mathbb{F}_p).$$

[Hint: Write the multiplication in the abelian group N additively and hence view N as a vector space over $\mathbb{F}_p$.]

Deduce that $\text{Aut } C_p \cong C_{p-1}$.

2. Let $G = \langle x \rangle$ be a cyclic group. Show that $\text{Aut } G$ is abelian.
3. Let N and H be groups, let $\phi: H \rightarrow \text{Aut } N$ be a homomorphism and $G = N \rtimes_{\phi} H$ the semidirect product constructed using these ingredients.
 - (a) Show that $h \mapsto (1, h)$ is a homomorphism $H \rightarrow G$ and that $\bar{H} = \{ (1, h) \mid h \in H \}$ is a subgroup of G that is isomorphic to H .
 - (b) Show that $\pi: (n, h) \mapsto h$ is a homomorphism $G \rightarrow H$. Determine the kernel of π .
Show that $\bar{N} = \{ (n, 1) \mid n \in N \}$ is a normal subgroup of G that is isomorphic to N and $G/\bar{N} \cong H$.
 - (c) Show that $G = \bar{N}\bar{H}$ and $\bar{H} \cap \bar{N} = \mathbf{1}$.
 - (d) Show that $(1, h)^{-1}(n, 1)(1, h) = (n^{h\phi}, 1)$ for all $h \in H$ and $n \in N$.
4. Show that the dihedral group D_{2n} is isomorphic to a semidirect product of a cyclic group of order n by a cyclic group of order 2. What is the associated homomorphism $\phi: C_2 \rightarrow \text{Aut } C_n$?
5. Show that the quaternion group Q_8 may not be decomposed (in a non-trivial way) as a semidirect product.
[Hint: How many elements of order 2 does Q_8 contain?]
6. Show that the symmetric group S_4 of degree 4 is isomorphic to a semidirect product of the Klein 4-group V_4 by the symmetric group S_3 of degree 3.
Show that S_4 is also isomorphic to a semidirect product of the alternating group A_4 by a cyclic group of order 2.

7. Let $G = F \rtimes T$ be the semidirect product of a cyclic group $F = \langle x \rangle$ of order 5 by $T = \langle y, z \rangle \cong C_2 \times C_2$ where $y^{-1}xy = x$ and $z^{-1}xz = x^4$.

Show that G is isomorphic to the dihedral group D_{20} of order 20.

[Hint: What is the order of the element xy ?]

8. Let G be a group of order pq where p and q are primes with $p < q$.
- (a) If p does not divide $q - 1$, show that $G \cong C_{pq}$, the cyclic group of order pq .
 - (b) If p does divide $q - 1$, show that there are essentially two different groups of order pq .

9. Classify the groups of order 52 up to isomorphism.

10. Show that a group of order 30 is isomorphic to one of

$$C_{30}, \quad C_3 \times D_{10}, \quad D_6 \times C_5, \quad D_{30}.$$

11. Classify the groups of order 98 up to isomorphism.

12. Classify the groups of order 117 up to isomorphism.

13. Let H and N be groups and let $\phi, \psi: H \rightarrow \text{Aut } N$ be two homomorphisms. Suppose that there exists $\alpha \in \text{Aut } N$ and $\beta \in \text{Aut } H$ such that $(h^\beta)\psi = \alpha^{-1}(h\phi)\alpha$ for all $h \in H$. Show that $(n, h) \mapsto (n^\alpha, h^\beta)$ is an isomorphism from $N \rtimes_\phi H$ to $N \rtimes_\psi H$.

[In this question, we are writing x^γ for the image of an element x of a group under an automorphism γ of the group. This is consistent with our notation in the semidirect product where we write $n^{h\phi}$ for the image of $n \in N$ under the automorphism $h\phi$.]

14. Let G and H be groups where G acts on the set $\Omega = \{1, 2, \dots, n\}$. Consider the wreath product $W = H \text{ wr}_\Omega G$ determined by this action and let B be the base group of W .

We shall view G as a subgroup of W by identifying it with the usual choice of complement to B naturally occurring in this semidirect product.

- (a) If N is a normal subgroup of H , show that the direct product $N \times N \times \dots \times N$ of n copies of N is a normal subgroup of W .
- (b) Show that an element $(h_1, h_2, \dots, h_n)$ of B commutes with all elements of G if and only if $h_i = h_j$ whenever i and j lie in the same orbit of G on Ω .
- (c) Suppose that $H \neq 1$ and that G acts transitively and also *faithfully* on Ω ; that is, the kernel of the associated permutation representation is trivial. Show that the centre of W is

$$Z(W) = \{ (h, h, \dots, h) \mid h \in Z(H) \}.$$

- (d) Suppose that $H \neq 1$ and that G acts faithfully on Ω . If N is a non-trivial normal subgroup of W , show that $N \cap B \neq 1$.

[Hint: If $N \cap B = 1$, what does this say about commutators $[x, b]$ where $x \in N$ and $b \in B$?]

15. Let V be a vector space over a field F . For emphasis below, we write $(V, +)$ for V viewed as a group with respect to the addition operation on the vector space.
- (a) Show that $\text{GL}(V)$ is a subgroup of $\text{Aut}(V, +)$.
 - (b) Deduce that we can construct the semidirect product $G = V \rtimes \text{GL}(V)$ via the natural action of $\text{GL}(V)$ on $(V, +)$.
 - (c) Show that V is a minimal normal subgroup of the semidirect product $G = V \rtimes \text{GL}(V)$.
 - (d) Show that an element x of G centralizes V (that is, commutes with all elements of V) if and only if $x \in V$.
 - (e) Deduce that V is the unique minimal normal subgroup of G .

[This semidirect product $V \rtimes \text{GL}(V)$ is called the *affine general linear group* and denoted by $\text{AGL}(V)$. It will return later in the lecture course.]